1° homogeneous linear system with constant coefficients

$x' = Ax$, let $x = \xi e^{rt}$

$\Rightarrow r\xi e^{rt} = A\xi e^{rt} \Rightarrow (A - rI)\xi = 0$

$\Rightarrow x = c_1 e^{\lambda_1 t} \cdot v_{\lambda_1} + c_2 e^{\lambda_2 t} v_{\lambda_2}$

e.g. $x' = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} x$

$\det(A - I\lambda) = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} = (\lambda-4)(\lambda-3) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda-2)(\lambda-5) = 0$

$\lambda = 2, 5$

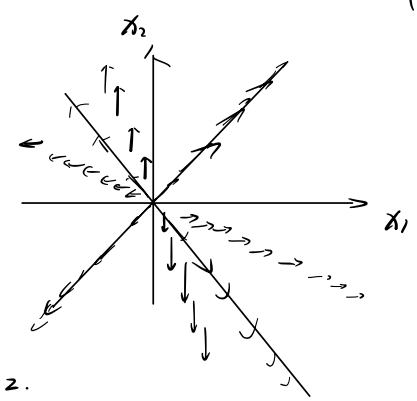
<1> $\lambda = 2$,

$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \to \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \to v_{\lambda_1} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

<2> $\lambda = 5$,

$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \to \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \to v_{\lambda_2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

s.t $x = c_1 e^{2t} \cdot \begin{pmatrix} -1 \\ 2 \end{pmatrix} + c_2 \cdot e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$



$\begin{cases} x_1' = 4x_1 + x_2 \\ x_2' = 2x_1 + 3x_2 \end{cases}$

1° $x_1' = 0$, $\Rightarrow 4x_1 + x_2 = 0$, $x_2 = -4x_1$

2° $x_2' = 0 \Rightarrow 2x_1 + 3x_2 = 0 \Rightarrow x_2 = -\frac{2}{3}x_1$

2.

Find the general solution of

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x}.$$

$\lambda \quad -\lambda^2 \quad \lambda$

$(1-\lambda)(1+\lambda)$

$(1+\lambda) + (1-\lambda)(1+\lambda)$

$\det(A - I\lambda) = \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = \begin{vmatrix} 0 & 1-\lambda^2 & 1+\lambda \\ 1 & -\lambda & 1 \\ 0 & 1+\lambda & -\lambda-1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & (1+\lambda)(2-\lambda) \\ 1 & -\lambda & 1 \\ 0 & 1+\lambda & -\lambda-1 \end{vmatrix}$

$$= \begin{vmatrix} 1 & -\lambda & 1 \\ 0 & 1+\lambda & -\lambda-1 \\ 0 & 0 & (1+\lambda)(2-\lambda) \end{vmatrix} = (1+\lambda)^2 \cdot (2-\lambda) = 0 \quad \Rightarrow \lambda = 2 \text{ or } -1 \text{ or } -1$$

<1> $\lambda = 2$

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

<2> $\lambda = -1$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$x_1 + x_2 + x_3 = 0$$
$$\Downarrow$$
$$0$$

2° complex-valued eigenvalues